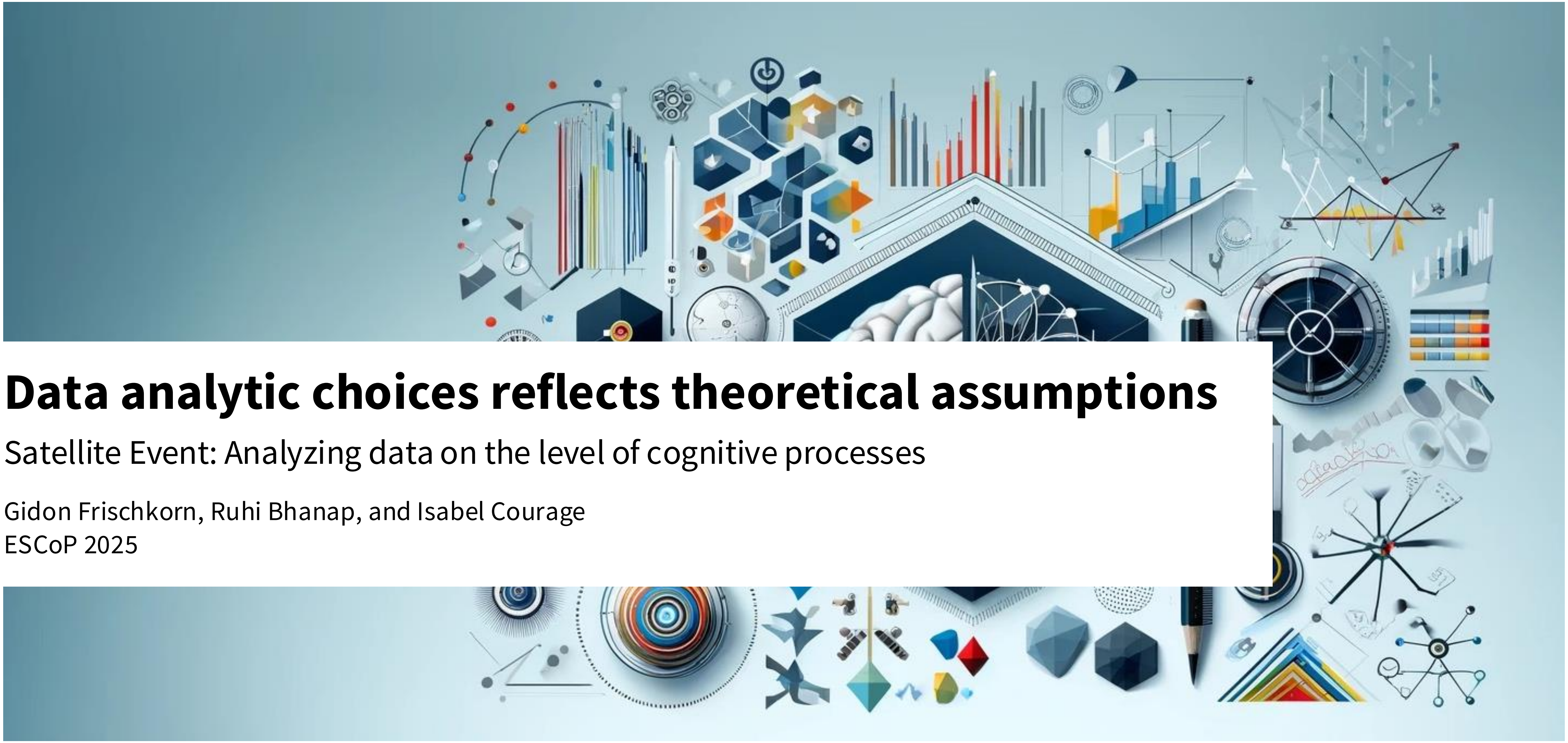
Universität Zürich[UZH]

Institute of Psychology



Data analytic choices reflects theoretical assumptions

Satellite Event: Analyzing data on the level of cognitive processes

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ESCoP 2025

Who are we?

PostDoc & SNF Ambizione Fellow Zurich

Mathematical Modelling of Cognition

Individual Differences in cognitive processes

PhD student University of Zurich

Modelling of Cognition

Declarative and procedural WM

PostDoc TU Dresden

Cognitive Models of Attention & Eye Movements

Visual attention and memory

Workshop: Overview

Monday

Morning

- Intro
  Why hierarchical models?
- Signal-Detection Models (brms)

Afternoon

- Memory Measurement Model (bmm)
- Practical Exercises

Tuesday

Morning

- Diffusion Decision Model (bmm)
- Practical Exercises

Afternoon

- Signal Discrimination Model
- Reporting Results from Cognitive Modelling Analyses

Materials are available via GitHub

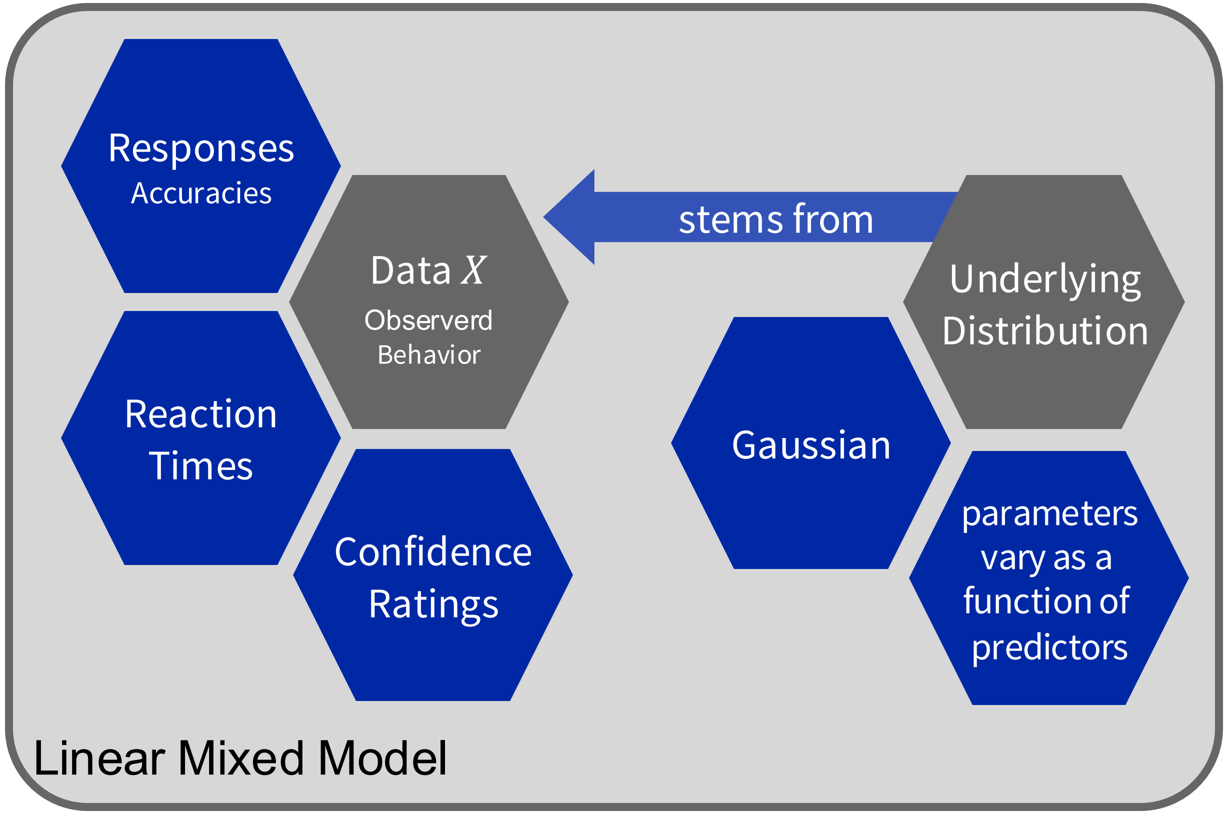
30 Min. Coffee Breaks
10:30 & 15:00

60 Min. Lunch Break
12:30

Ask questions whenever something is not clear
☺

Data Analytic Model

i = participants
j = trials
k = conditions



$$X \quad \sim D(\theta)$$

D = Symbol for a specific probability distribution
θ = parameters of distributions

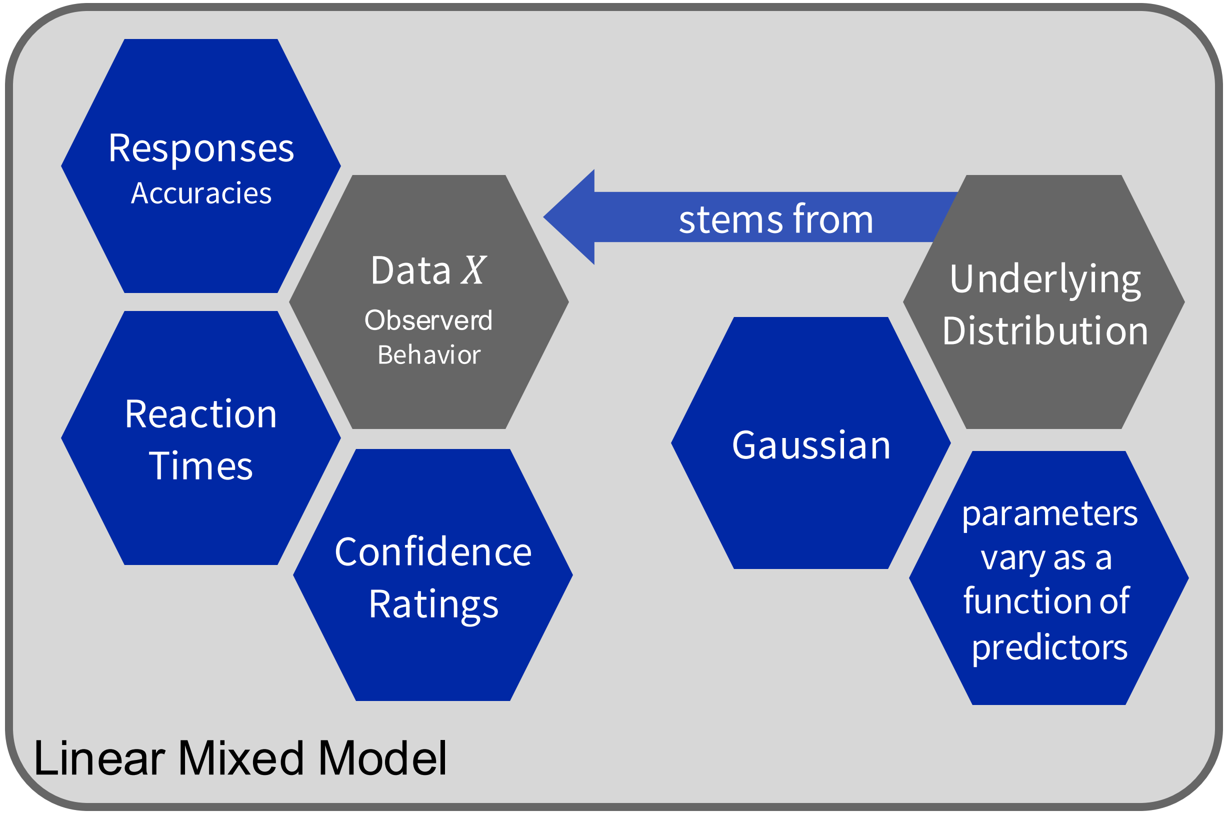
$$\theta = \beta_0 + \sum_{k=1}^{K} \beta_k * Y_k$$

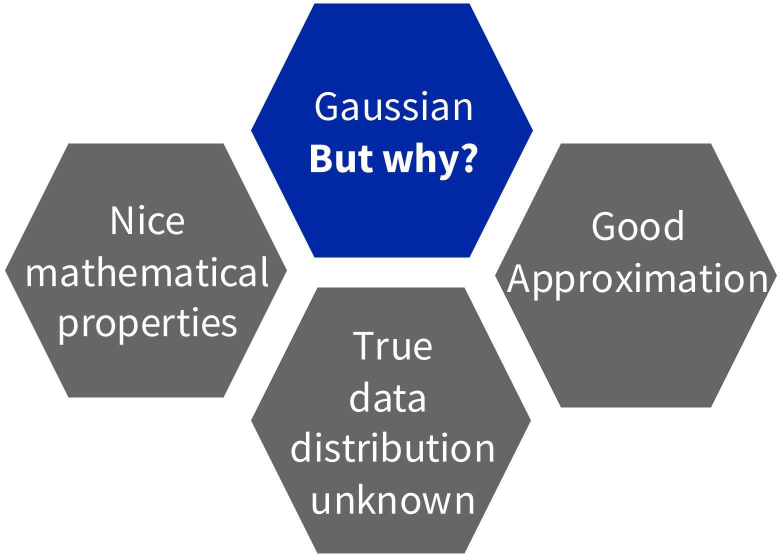
Linear combination of:
β_0 = Intercept → the parameter value if all predictors are zero
$\beta_k * Y_k$ = a group of K predictors Y weighted by β_k

Data Analytic Model

Intro

Why hierarchical models?





The Central Limit Theorem

Intro

Why hierarchical models?

Excercise I

1. Sample 2, 5, 10, and 20 values from a sample of values 1 to 5
2. Choose arbitrary probabilities for each of the five values
3. Compute the mean of the samples

Repeat step 1 to 3 for 200 times and plot the means for all 200 iterations

Mean is smaller then infinity

Variance Is smaller then infinity

X ~ D
D = any Probability distribution

Central Limit Theorem

i indepen-dent samples of X

$\sum X_i \sim$ *Normal*

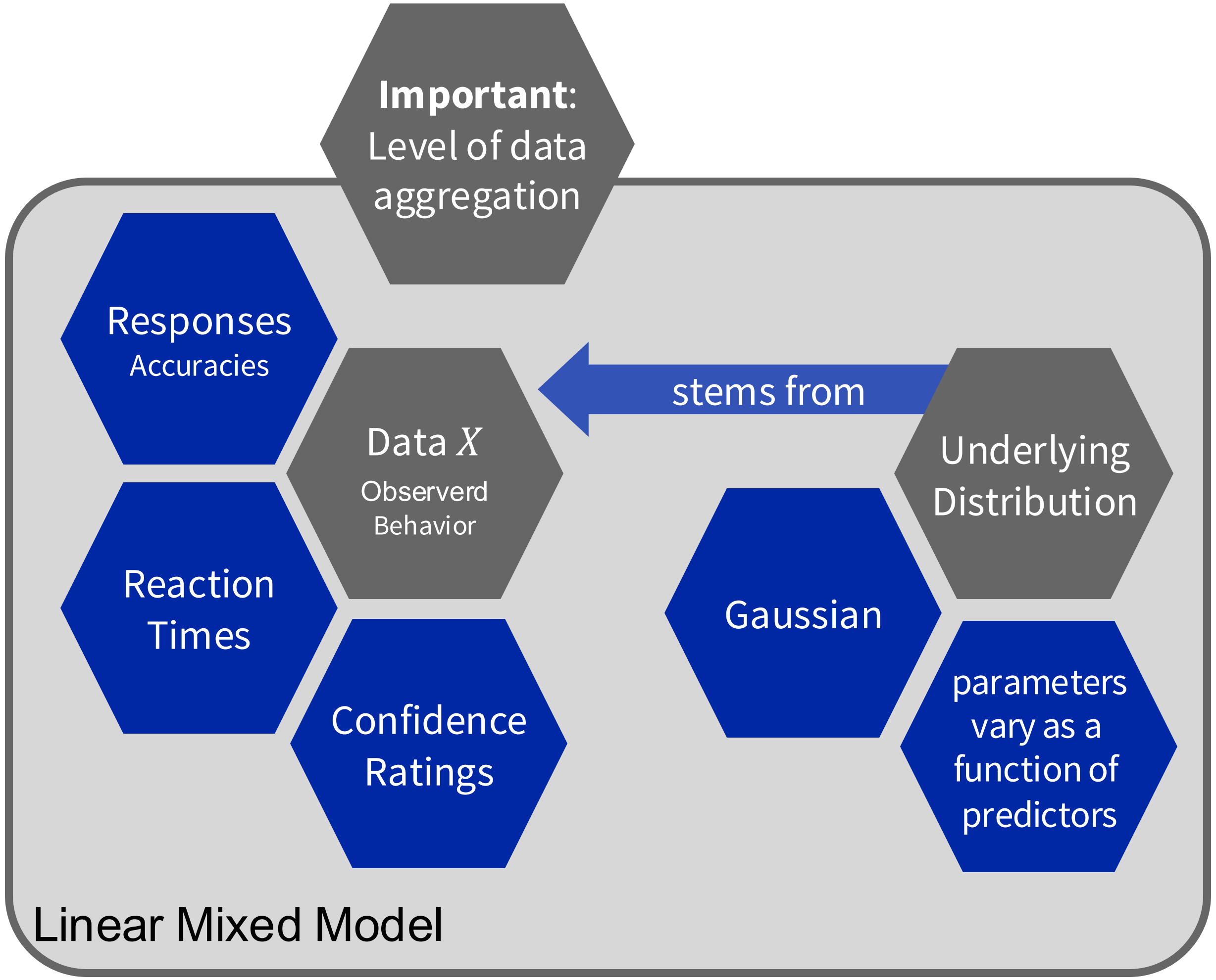
Gaussian
But why?

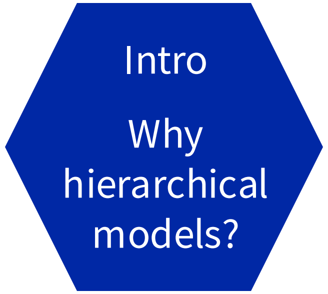
Nice mathematical properties

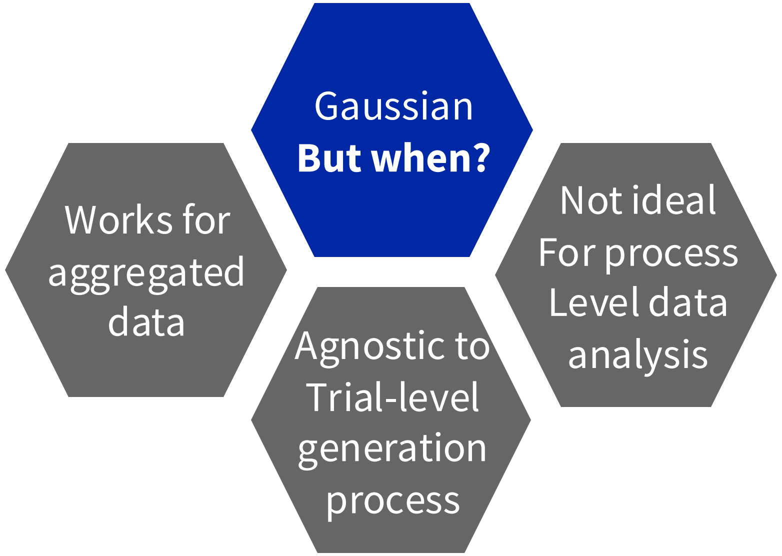
True data distribution unknown

Good Approximation

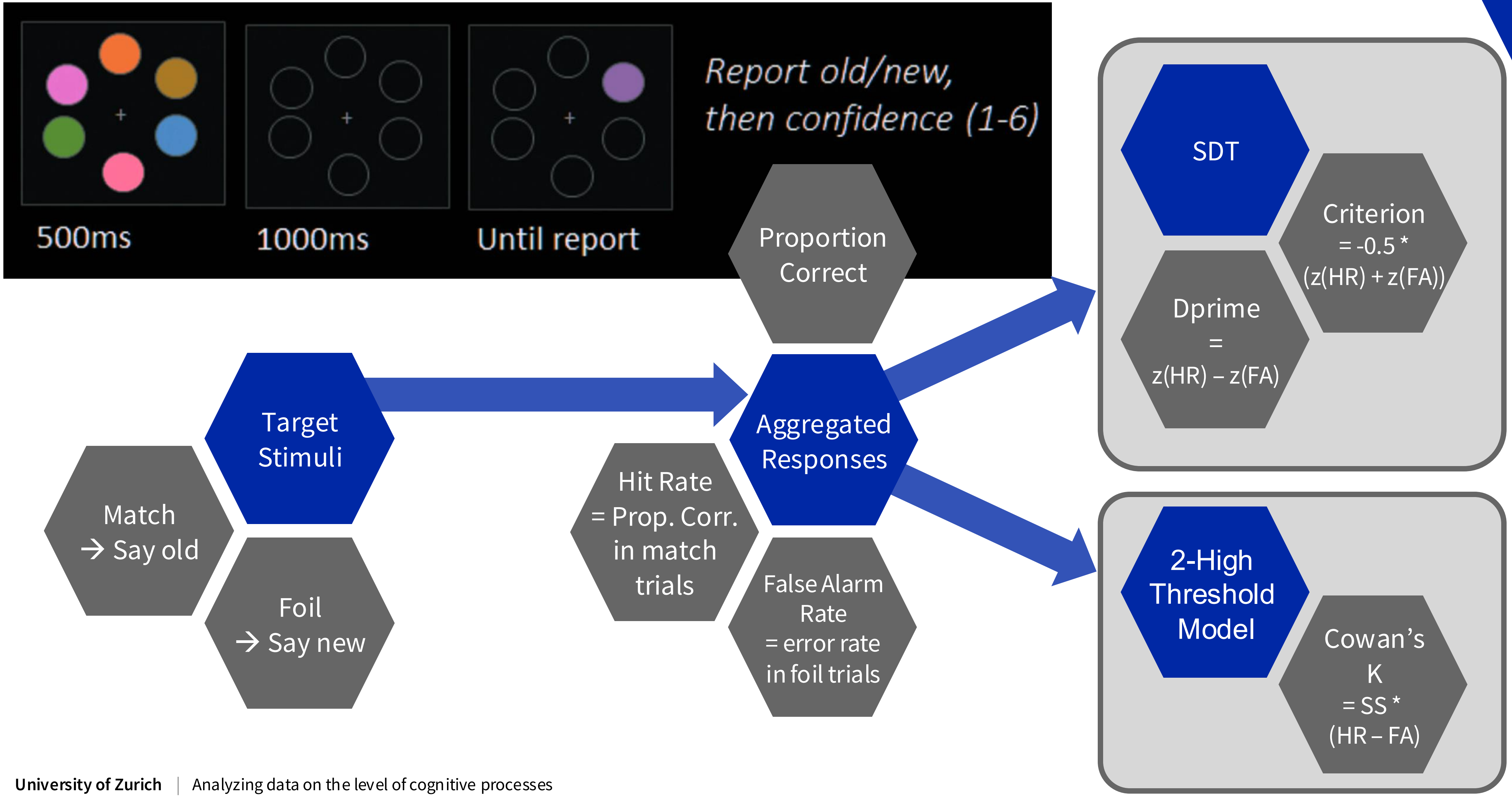
Data Analytic Model



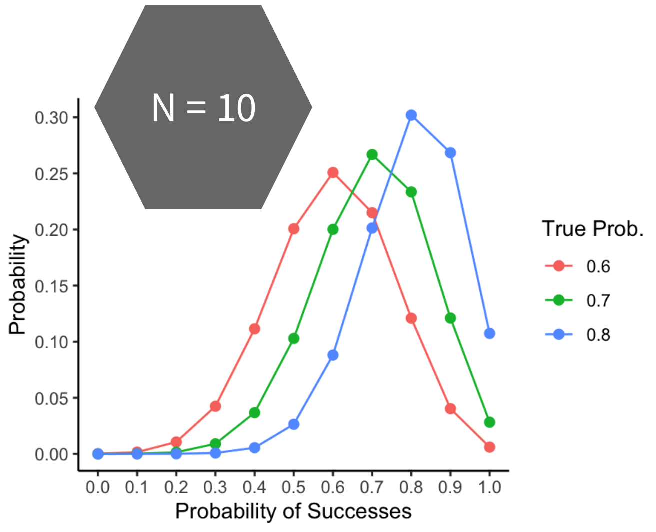


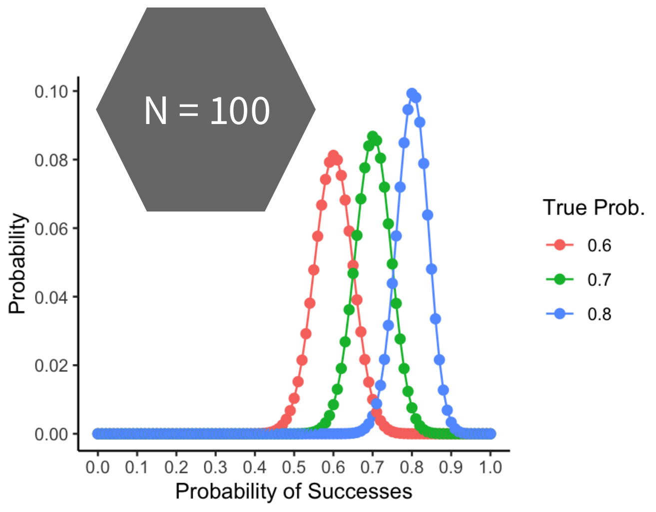


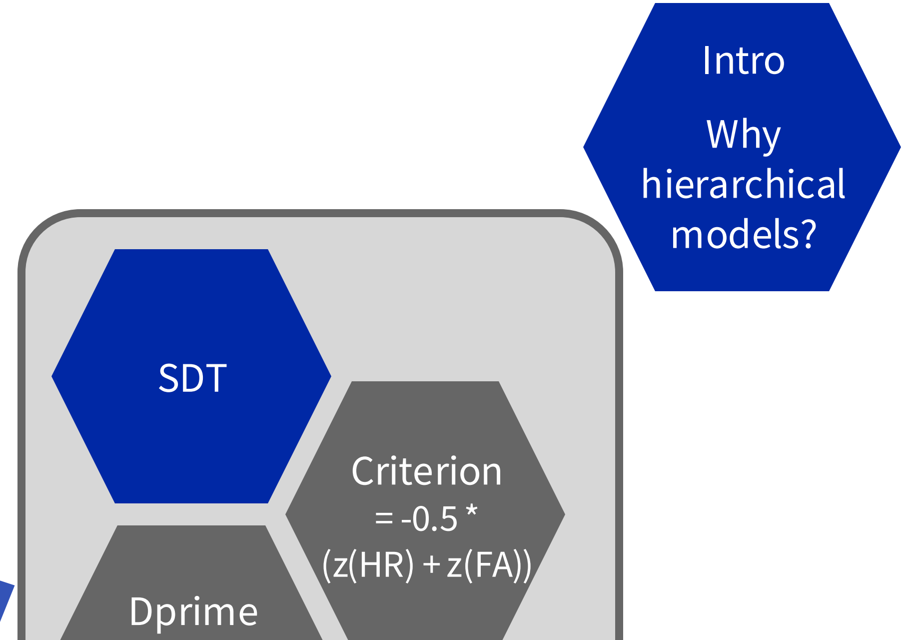
How to aggregate our data?

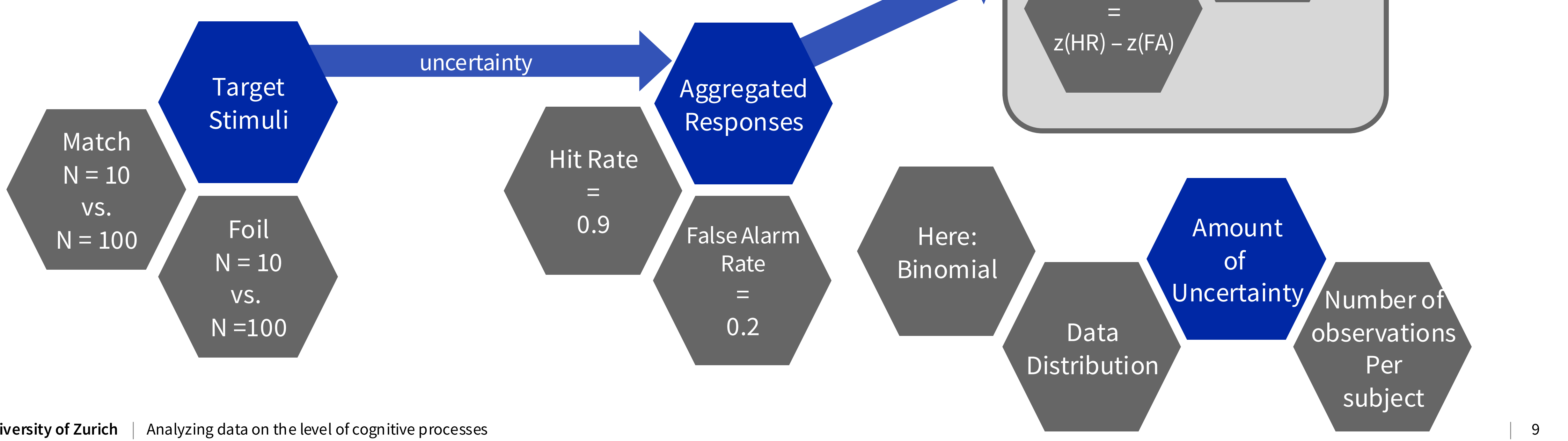


Uncertainty in aggregated data

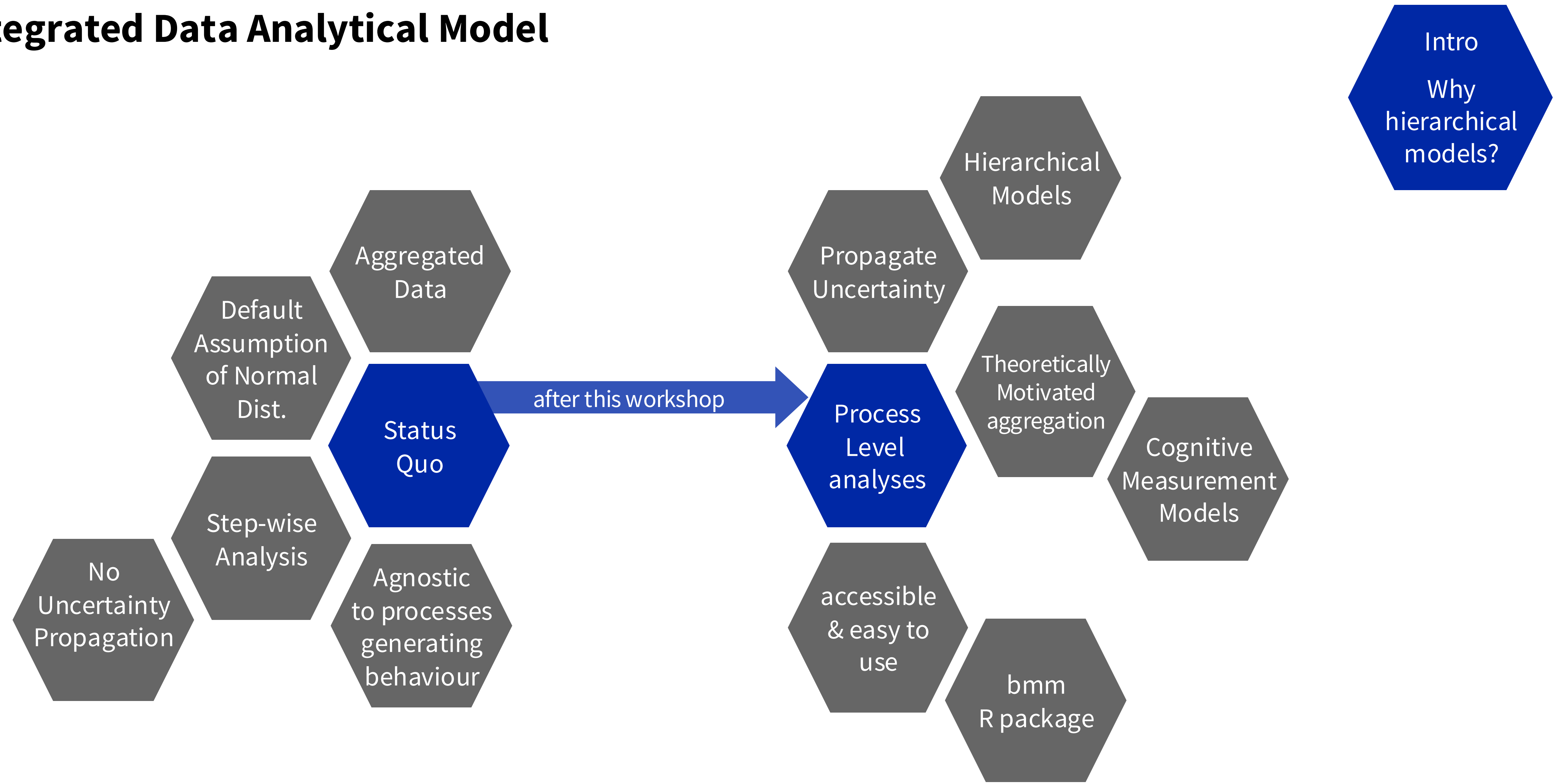








Integrated Data Analytical Model



What are your questions so far?

A short introduction to Bayesian statistics

Bayesian Logic

Prior Belief × Data = Posterior Belief

2. data

1. prior belief about hypothesis

3. update belief about hypothesis

$$P(H|D) = \frac{P(H) * P(D|H)}{P(D)}$$

$$P(D) = P(H) * P(D|H) + P(\neg H) * P(D|\neg H)$$

Prior = probability of a given a hypothesis or model

Likelihood = probability of data given a hypothesis or model

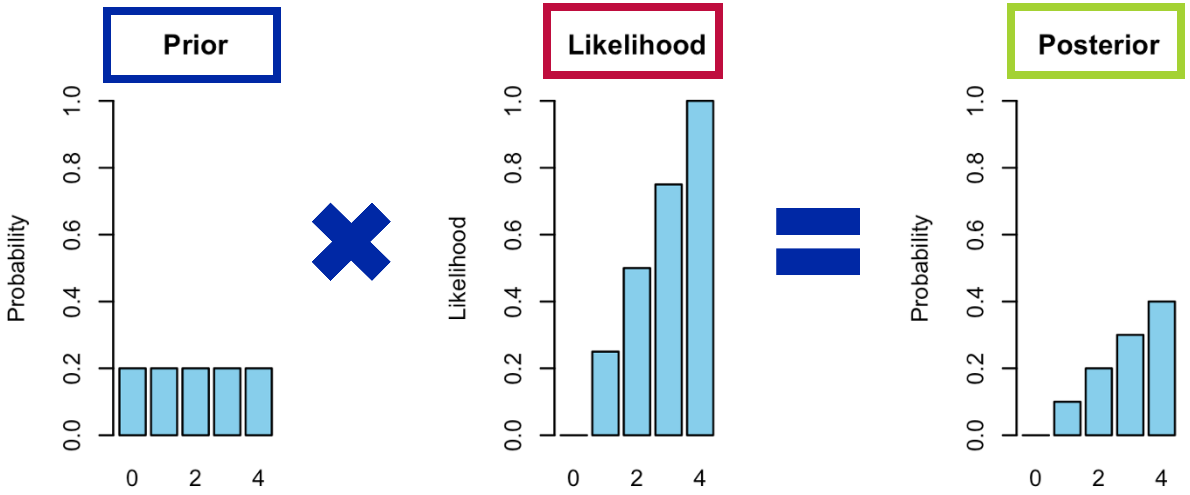
Bayes Theorem

Posterior = Probability Of hypothesis/ Model given the data

Marginal Likelihood = probability of the data

A short introduction to Bayesian statistics

Bayesian Logic



Prior
= probability
of a given
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Likelihood
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Bayes
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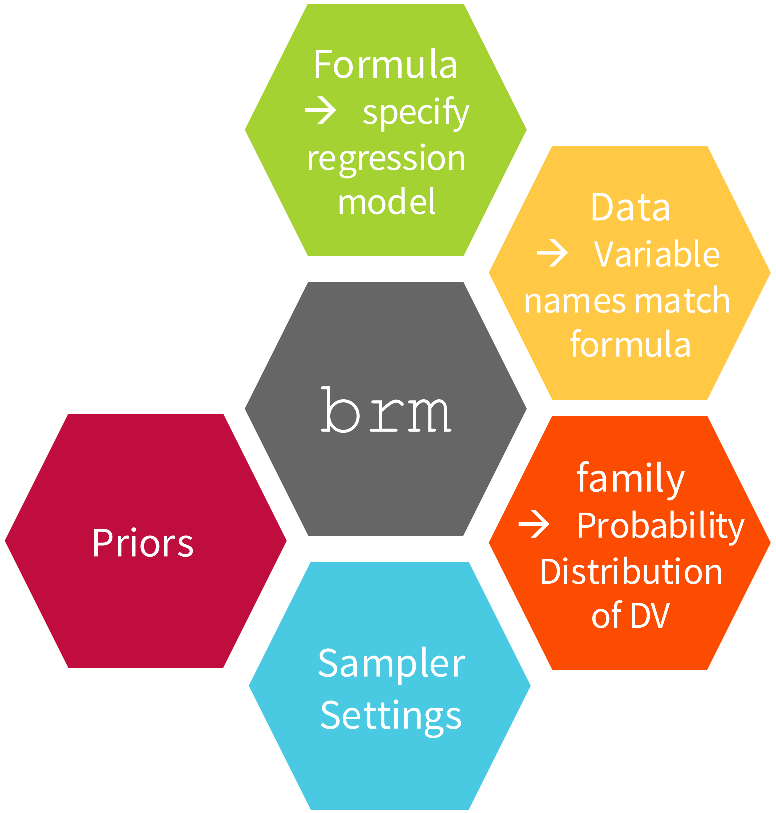
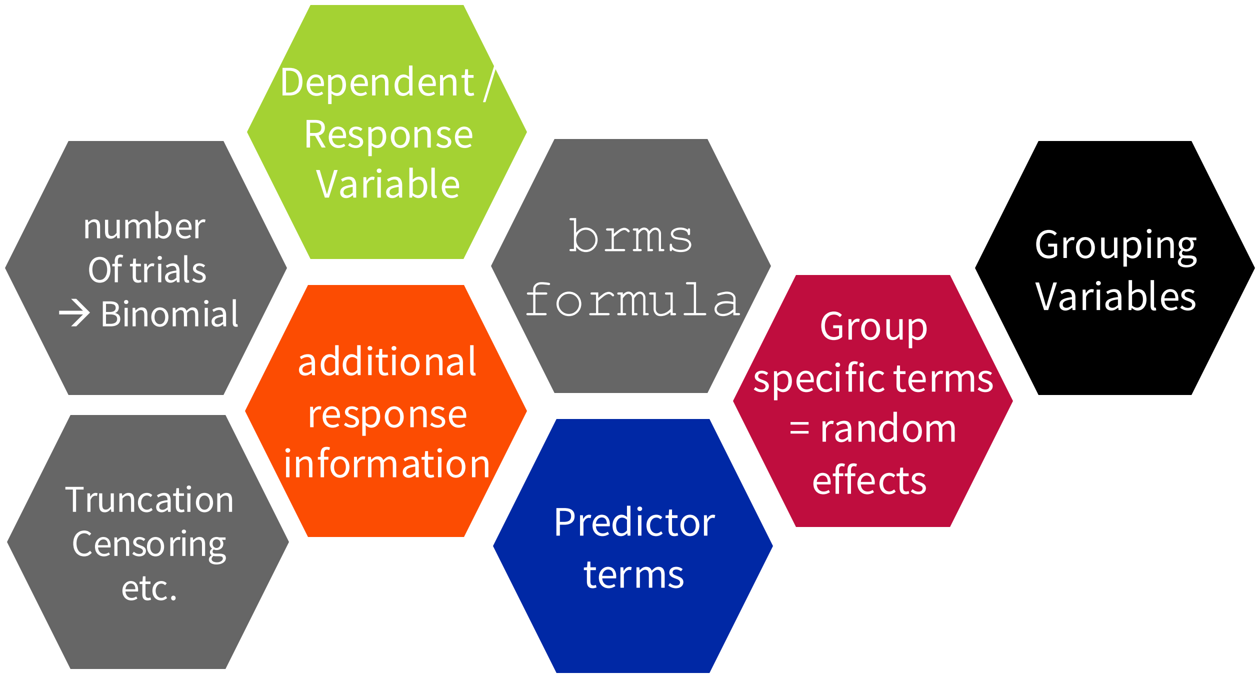
The brms R package




```
brm(formula = DV ~ 1 + IV + (1 + IV | ID),
    data = myData,
    # optional arguments
    family = gaussian(),
    prior = myPriors,
    iter = 4000, warmup = 1500,
    nChains = 4)
```

The brms R package

```
response | aterms ~ pterms + (gterms | group)
         dist_parm ~ pterms + (gterms | group)
```



Important: By default, this formula predicts the primary distribution parameter → mean, location, or probability parameters

→ You can specify additional formulae to predict other distributional parameters (e.g. sigma)

The brms R package

```
default_prior(formula, data, family)
```

default_prior

Formula → specify regression model

Data → Variable names match formula

family → Probability Distribution of DV

prior

Distribution + parameters

Model Parameter class

```
my_prior <- prior(normal(0,1), class = b) +
    prior(gamma(1,1), class = sigma, lb = 0)
```

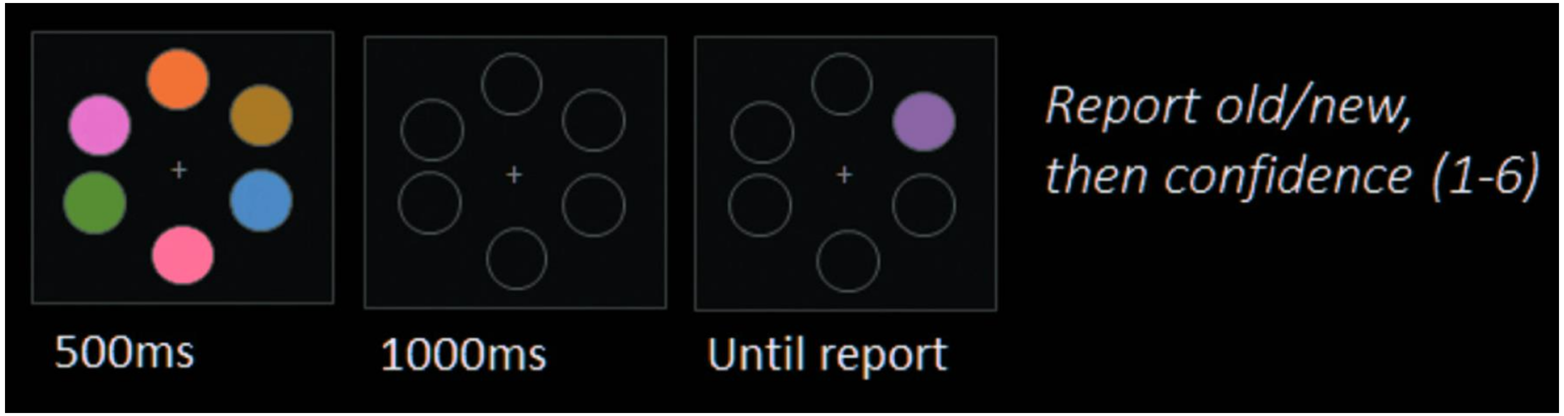
```
> default_prior(formula, ToothGrowth)
                prior     class        coef group resp dpar nlpar lb ub       source
               (flat)         b                                              default
               (flat)         b       dose1                             (vectorized)
               (flat)         b       dose2                             (vectorized)
               (flat)         b       supp1                             (vectorized)
               (flat)         b supp1:dose1                             (vectorized)
               (flat)         b supp1:dose2                             (vectorized)
 student_t(3, 19.2, 9) Intercept                                             default
     student_t(3, 0, 9)     sigma                                  0           default
```

for more info see the documentation.

What are your questions so far?

The brms R package

Exercise II A



Aggregated Responses

Proportion Correct

Cowan's K = SS * (HR – FA)

Dprime = z(HR) – z(FA)

Experimental Manipulation

Adaptive Instruction → Answer old & new equally

Excercise II A

1. Load the Williams et al. (2022) aggregated data
2. Specify a between group model predicting one DV by the isAdaptive variable
3. Fit the model, and try to conclude if there is an effect of the manipulation